

Compute Cost Pyramid

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Training vs. inference vs. fine-tune

PROFESSIONAL DEFINITION

The Compute Cost Pyramid distinguishes the cost profiles of training, fine-tuning and inference, showing that while training is rare and expensive, inference is continuous and scales with usage.

WHO DEVELOPED IT

A practitioner construct from machine-learning operations and cloud-economics analysis of large-model lifecycles.

WHY IT WAS DEVELOPED

It was developed to correct the assumption that training dominates AI cost; for deployed systems, inference at scale is frequently the larger bill.

FIGURE 1 · COMPUTE COST PYRAMID

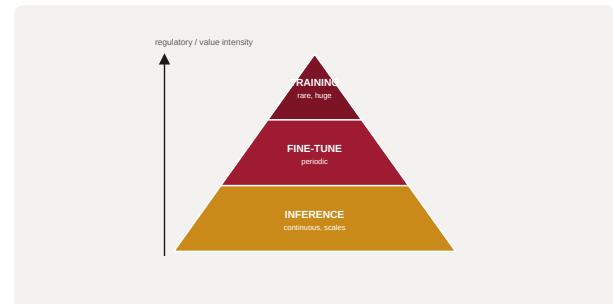


Figure 1. A schematic of the Compute Cost Pyramid as applied in BARBM312 (illustrative).

HOW IT IS USED

Costs are allocated across training, fine-tuning and inference; planning focuses on the layer that dominates the firm's actual usage pattern.

WHEN IT IS USED

During AI cost forecasting and architecture decisions.

BY WHOM IT IS USED

ML engineering, finance and platform teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It reframes AI cost around lifecycle stage, highlighting inference economics that one-off training-cost framing obscures.

RELATED FRAMEWORKS IN THIS COURSE

AI Value Stack · Slibrary 9

Token Economics · Slibrary 9

Hyperscaler Dependency Map · Slibrary 9

Capex vs. Opex Shift · Slibrary 9

SOURCES (HARVARD REFERENCING STYLE)

Patterson, D. et al. (2021) 'Carbon emissions and large neural network training', arXiv:2104.10350.

Sevilla, J. et al. (2022) 'Compute trends across three eras of machine learning', IJCNN.